1 Performance Benchmarks

Performance Comparison Report

Advanced Framework Benchmarks across CPU and GPU Architectures

1.1 LAMMPS vs. gpu_mc (Large-Scale Scales)

Benchmarks were conducted using two distinct NVIDIA hardware architectures: the **RTX 2000** and the **RTX 4500**.

Table 1: Simulation Runtime (Seconds) by Particle Count (N) for Large Scales

	LAMMPS Time		**gpu_mc Time**	
Particle Count (N)	**RTX 2000**	**RTX 4500**	**RTX 2000**	**RTX 4500**
492 450	7.0	5.0	18.0	3.8
1 969 920	22.0	18.0	66.0	13.2
3 939 840	44.0	34.0	123.0	25.6

1.2 gpMC (CPU) vs. mc_gpu (GPU) (Small-to-Medium Scales)

This benchmark scales from 8000 to approximately 110 000 particles, comparing a CPU-driven implementation (**gpMC**) against dedicated GPU acceleration (**mc_gpu**).

For the CPU execution on **gpMC**, timings are provided for the Core **i7-14700K**. Testing on the **Intel Xeon Gold 6548Y+** demonstrated an approximate relative performance factor of 0.98× (i.e., 1.02× slower execution times across workloads).

2 Discussion and Structural Insights

An analysis of the benchmarking profiles yields two key architectural insights:

Table 2: Simulation Runtime (Seconds) across CPU and GPU Architectures

	gpMC (CPU)	**mc_gpu (GPU) Time**	
Particle Count (N)	**i7-14700K†**	**RTX 2000**	**RTX 4500**
8000	10.0	1.21	0.85
15 626	23.4	4.27	3.20
32 768	38.4	3.49	2.61
125 000	167.0	4.34	1.93

† For Intel Xeon Gold 6548Y+, scale execution times by a factor of ~ 1.02 ($0.98\times$ slower).

- **Heterogeneous Paradigm Shifts:** Moving from multicore CPU environments (`gpMC`) to even entry-level professional GPUs (`mc_gpu` on RTX 2000) yields over an order of magnitude speedup (e.g., $27\times$ speedup at the largest medium scale).

- **Anomalous Scaling Profile:** In the `mc_gpu` dataset on the RTX 2000/4500 architectures, scaling from $N = 15\,626$ to $N = 32\,768$ demonstrates an absolute reduction in execution time. This indicates a hardware architectural threshold where GPU thread occupancy optimizations and block-level memory coalescing align to hit peak efficiency.